

txt2tex Reference

Write math like you're at a whiteboard. Get L^AT_EX you can hand in.

<https://github.com/jmf-pobox/txt2tex>

Document Structure

Single-line directives.

=== Title ===	<code>\section *{Title}</code>
** Solution N **	Bold heading with spacing
(a) Text	Part label
TITLE: My Doc	Document title (<code>\title {}</code>)
SUBTITLE: Sub	Document subtitle
AUTHOR: Name	Author name
DATE: 2026	Date string
INSTITUTION: Univ	Institution line
BIBLIOGRAPHY: f.bib	BibTeX file
BIBLIOGRAPHY_STYLE: plainnat	Citation style
CONTENTS:	Table of contents
PARTS: inline	Parts formatting style
PAGEBREAK:	<code>\newpage</code>
LINEBREAK:	<code>\medskip</code> (gap, no new page)
TEXT: prose here	Smart prose (auto-detects formulas)
PURETEXT: raw & text	Verbatim prose; escapes L ^A T _E X specials
LATEX: <code>\cmd</code>	Raw L ^A T _E X passthrough (single-line)
[cite key]	<code>\citep {key}</code> (in TEXT blocks)

END-terminated blocks.

LATEX:	Multi-line raw L ^A T _E X block; body until END
B:	B-machine verbatim block; body until END

L^AT_EX: with content on the same line is single-line form. L^AT_EX: alone on a line opens the block form — the body is live L^AT_EX. B: wraps its body in `verbatim` — backslashes print as typed. Both require a bare END at column 0 to close.

You write:

```
LATEX:
\begin{tikzpicture}
  \draw (0,0) -- (1,1);
\end{tikzpicture}
END
```

You write:

```
B:
MACHINE Counter
VARIABLES n
INVARIANT n : NAT
END
```

Identifier Decoration.

<code>count'</code>	<i>count'</i> (after-state)
<code>in?</code>	<i>in?</i> (input)
<code>out!</code>	<i>out!</i> (output)
<code>x?'</code>	<i>x?'</i> (combinations allowed)
<code>'sunk'</code>	'sunk' (string literal)

Decoration attaches to any identifier. Reserved words (`theta`, `defs`, `hide`, `project`, `Delta`, `Xi`) cannot be decorated.

Propositional Logic

<code>p land q</code>	$p \wedge q$
<code>p lor q</code>	$p \vee q$
<code>lnot p</code>	$\neg p$
<code>p => q</code>	$p \Rightarrow q$
<code>p <=> q</code>	$p \Leftrightarrow q$

Note: Use `land`, `lor`, `lnot`. English `and/or/not` are not supported.

Truth-table example:

You write:

```
TRUTH TABLE:
p | q | p => q
T | T | T
T | F | F
F | T | T
F | F | T
```

You get:

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Equivalence chain (EQUIV:):

You write:

```
EQUIV:
lnot (p land q)
lnot p lor lnot q  [De Morgan]
```

You get:

$$\neg (p \wedge q)$$

$$\Leftrightarrow \neg p \vee \neg q$$

[De Morgan]

EQUIV: and its alias **ARGUE:** join steps with $\Leftrightarrow$. Use for predicate equivalences.

Equality chain (EQUAL:):

You write:

EQUAL:

0 + 0

0

length(<>)

[0+0 = 0]

[length(<>) = 0]

You get:

0 + 0

= 0

= *length*(⟨⟩)

[0+0 = 0]

[length(⟨⟩) = 0]

EQUAL: joins steps with =. Use when steps are arithmetic/set-valued, not propositional equivalences.

Numbers & Arithmetic

N	$\mathbb{N}$ (natural numbers)
Z	$\mathbb{Z}$ (integers)
N1	$\mathbb{N}_1$ (positive naturals)
n + m / n - m	addition / subtraction
n * m / n div m	multiplication / integer division
n mod m	modulo
n < m / n > m	strict ordering
n <= m / n >= m	non-strict ordering
n = m / n /= m	equality / inequality

Exponentiation: $\mathbf{R}^{\mathbf{n}}$ (no space) emits `\bsup n\esup` — the fuzz *iter* operator. It is only valid for relations, not arithmetic. For x^2 write `x * x`.

Sets & Types

x elem S	$x \in S$
x notin S	$x \notin S$
A subset B	$A \subseteq B$
A psubset B	$A \subset B$ (proper subset)
A union B	$A \cup B$
A intersect B	$A \cap B$
A setminus B	$A \setminus B$
bigcup S	$\bigcup S$
bigcap S	$\bigcap S$
power S	$\mathbb{P} S$
P1 S	$\mathbb{P}_1 S$ (non-empty)
F S	$\mathbb{F} S$
F1 S	$\mathbb{F}_1 S$ (non-empty finite)
A cross B	$A \times B$
{x : S P}	Set comprehension
n..m	$n..m$ (integer range)
# S	$\#S$ (cardinality / length)

Prefix-operator rule (#133): prefix-generic operators (seq, P, dom, ran, bigcup, ...) require an atomic argument. The engine automatically wraps a nested prefix call in parentheses: seq (P X) → \seq~(\power X).

Set comprehension:

You write:

```
{x : N | x > 0 land x < 5}
```

You get:

$$\{x : \mathbb{N} \mid x > 0 \wedge x < 5\}$$

Predicate Logic

forall x : S P	$\forall x : S \bullet P$
exists x : S P	$\exists x : S \bullet P$
exists_1 x : S P	$\exists_1 x : S \bullet P$ (unique)
lambda x : S E	$\lambda x : S \bullet E$
mu x : S P	$\mu x : S \bullet P$ (definite desc.)
if p then e1 else e2	conditional expression

Bullet form: x : S | P means $x : S \bullet P$ (bullet is implied). The vertical bar is the quantifier separator.

Tuple-pattern:

You write:

```
forall (x, y) : N cross N |
  x + y >= 0
```

You get:

$$\forall (x, y) : \mathbb{N} \times \mathbb{N} \bullet x + y \geq 0$$

Multi-decl: separate variable groups with ; inside a quantifier: forall x : A; y : B | P.

Schema-text scope: a bare [d : T | P] in a quantifier or function acts as an inline schema text.

Relations

A <-> B	$A \leftrightarrow B$ (relation type)
x -> y	$(x \mapsto y)$ (maplet)
dom R / ran R	dom R / ran R
id	identity relation
inv R / R~	R^{-1} (inverse; postfix or prefix)
A dres R / A < R	$A \triangleleft R$ (domain restrict)
A dsub R / A << R	$A \triangleleft\!\!\!\triangleleft R$ (domain subtract)
R rres B / R > B	$R \triangleright B$ (range restrict)
R rsub B / R » B	$R \triangleright\!\!\!\triangleright B$ (range subtract)
R oplus S / R ++ S	$R \oplus S$ (relational override)
R o9 S	$R \circ_9 S$ (forward comp.)
R comp S	$R \circ_8 S$ (backward comp.)
R(A)	$R[A]$ (relational image)
shows	$\vdash$ (turnstile / judgment)

o9 vs comp: o9 → \semi (forward, $(R; S)(x) = S(R(x))$). comp → \comp (backward, $(R \circ S)(x) = R(S(x))$).

Worked example — domain restrict:

You write:

```
{1, 2} dres {1 |-> a, 2 |-> b, 3 |-> c}
```

You get:

$$\{1, 2\} \triangleleft \{1 \mapsto a, 2 \mapsto b, 3 \mapsto c\}$$

Functions

<code>A ++> B</code>	$A \leftrightarrow B$ (partial)
<code>A -> B</code>	$A \longrightarrow B$ (total)
<code>A >+> B / A - > B</code>	$A \rightsquigarrow B$ (partial injection)
<code>A >-> B</code>	$A \rightharpoonup B$ (total injection)
<code>A ++» B</code>	$A \twoheadrightarrow B$ (partial surjection)
<code>A -» B</code>	$A \twoheadrightarrow B$ (total surjection)
<code>A >-» B</code>	$A \twoheadrightarrow B$ (bijection)
<code>A 77-> B</code>	$A \dashrightarrow B$ (finite partial)
<code>f(x)</code>	function application
<code>lambda x : T f(x)</code>	$\lambda x : T \bullet f(x)$
<code>f ++ g</code>	$f \oplus g$ (override)
<code>f</code>	f^{-1} (postfix inverse)

Worked example — lambda:

You write:

```
double == lambda n : N | n * 2
```

You get:

double == $\lambda n : \mathbb{N} \bullet n * 2$

Sequences & Bags

<code><> / <a, b></code>	$\langle \rangle / \langle a, b \rangle$
<code>s ^ t</code>	$s \frown t$ (concat; space before ^ required)
<code>seq S / seq1 S</code>	$\text{seq } S / \text{seq}_1 S$
<code>iseq S</code>	$\text{iseq } S$ (injective sequences)
<code># s</code>	$\#s$ (length)
<code>head s / tail s</code>	first / rest
<code>last s / front s</code>	last / all-but-last
<code>rev s</code>	reverse
<code>s(i)</code>	index (1-based)
<code>s filter A</code>	$s \upharpoonright A$ (filter)
<code>bag S</code>	$\text{bag } S$
<code>[[a, b]]</code>	$\llbracket a, b \rrbracket$ (bag literal)
<code>b1 bag_union b2</code>	$b_1 \uplus b_2$
<code>b1 bag_diff b2</code>	$b_1 \ominus b_2$

Concat: write `<a> ^ <b>` (space before ^). No space $\rightarrow$ relation iteration (`\bsup`). **bag_diff:** emits `\uminus` — bag subtraction clamped at zero.

Z Paragraphs

given A, B Given types: $[A, B]$
T ::= a | b Free type definition
name == expr Abbreviation

axdef / **schema** Name / **gendef** [X]
 declarations Variable declarations
where Separator
 predicates Constraining predicates
end Close the block

axdef introduces global constants; **gendef** [X] parameterises by a type; **schema** Name names a state or operation.

Zed block (unboxed paragraph):

You write:

```
zed
  given Entry
  Status ::= active | inactive
  MaxSize == 100
end
```

You get:

```
[Entry]

Status ::= active | inactive

MaxSize == 100
```

zed blocks can mix given types, free types, abbreviations, and predicates in one environment. No visual box. Use for global type definitions; use **axdef**/**schema** for declarations with a **where** clause.

Syntax block (multi-type free definitions):

You write:

```
syntax
  OP ::= plus | minus | times
  Expr ::= const<N> | binop<OP cross Expr>
end
```

syntax produces a column-aligned fuzz **syntax** environment. Blank lines between definitions emit \also (visual group separator). Use for BNF-style grammars and type hierarchies.

Schemas

A named schema introduces a signature (declarations) and an invariant (where clause).

You write:

```
schema Counter
  size : N
  nums : seq N
where
  size = # nums
end
```

You get:

```
Counter
size : N
nums : seq N

size = #nums
```

Where-clause patterns:

1. Simple comparison.

You write:

```
schema Bounded
  size : N
where
  size < 10
end
```

2. Set comprehension.

You write:

```
schema Inventory
  items : seq N
  stock : N
where
  stock = # {x : ran items | x > 0}
end
```

3. Quantifier.

You write:

```
schema NonNeg
  s : seq Z
where
  forall i : 1..#s | s(i) >= 0
end
```

Generic schemas.

You write:

```
gendef [X]
  empty : F X
where
  empty = {}
end
```

You get:

$\langle X \rangle$ is the type parameter; instantiate with $G[T]$.

Schemas as predicates.

Inside a quantifier, a schema name acts as a predicate (both its signature and its where clause must be satisfied):

You write:

```
forall c : Counter |
  c.size >= 0
```

You get:

$$\forall c : Counter \bullet c.size \geq 0$$

Schema Operations

Inclusion operators.

SName	bare inclusion
Delta Counter	$\Delta Counter$ (before/after-state pair)
Xi Card	$\Xi Card$ (read-only state)
theta S	θS (binding of current state)
theta S'	$\theta S'$ (binding of after-state)

Horizontal definition.

Name	defs	RHS:
Name	defs	Schema
N	defs	Delta C
N	defs	[d : T P]
N[X]	defs	G[X]
		generic form

Schema calculus operators (RHS of defs).

S[old/new]	rename component
S[a/b, c/d]	multiple renames
S ; T	$S \circ T$ (composition)
S » T	$S \gg T$ (piping)
S hide (x, y)	$S \setminus (x, y)$
S project T	$S \upharpoonright T$

Z Bindings

Z RM §3.7: binding brackets construct labelled tuples.

{ name == e }	$\langle name == e \rangle$
{ a == e, b == f }	multiple components
{ }	empty binding

Multi-typed comprehension with binding:

You write:

```
{ t : Track; a : Album |
  t.albumId = a.albumId .
  { | trackId == t.trackId | } }
```

You get:

$$\{ t : Track; a : Album \mid t.albumId = a.albumId \\ \bullet \langle trackId == t.trackId \rangle \}$$

Use `;` to separate variable-type pairs inside `{ }`. Binding components are **comma**-separated (Z RM §3.7).

Fuzz note: bindings emit outside any Z environment. Do not place them inside `zed/axdef/schema` blocks — fuzz rejects `==` inside $\langle \dots \rangle$ in block contents.

Proofs

INFRULE: — named inference rule display:

You write:

```
INFRULE:
premise_1
premise_2
---
conclusion [Rule name]
```

You get:

$$\frac{\textit{premise}_1 \quad \textit{premise}_2}{\textit{conclusion}} \text{ Rule name}$$

INFRULE: renders a labelled rule schema. Use for displaying rule definitions (not proofs). The three-dash line `--` separates premises from conclusion.

How proof trees work (PROOF:):

The source is **conclusion-first** (rendered tree reads bottom-up). Premises are indented under their conclusion. A rule with two or more premises requires `::` on each premise — without it, indented lines collapse into a *linear chain* (each line becomes the sole premise of the line above) and the extra premises are silently dropped. Unary rules take a single indented child with no `::`.

Proof tree syntax:

indent	Premises indented under conclusion
[rule]	Justification label
[rule from <i>n</i>]	Discharge assumption <i>n</i>
[<i>n</i>] expr [assumption]	Boxed assumption labelled <i>n</i>
expr [from <i>n</i>]	Leaf reference to assumption <i>n</i>
::	Premise of a multi-premise rule
case <i>X</i> :	Branch in a lor elim case analysis

Rules: `land intro`, `land elim 1/2`, `lor intro 1/2`, `lor elim`, `=> intro/elim`, `<=> intro/elim`, `lnot intro/elim`, `forall intro/elim`, `exists intro/elim`.

Modus ponens (binary — requires `::` on each premise):

You write:

```
PROOF:
q [=> elim]
:: p => q
:: p
```

You get:

$$\frac{p \Rightarrow q \quad p}{q} \Rightarrow \text{-elim}$$

$\wedge$ -intro (binary):

You write:

```
PROOF:
p land q [land intro]
:: p
:: q
```

You get:

$$\frac{p \quad q}{p \wedge q} \wedge \text{-intro}$$

$\Rightarrow$ -intro with discharge (use from N):

You write:

```
PROOF:
p => p [=> intro from 1]
[1] p [assumption]
:: p [from 1]
```

You get:

$$\frac{\ulcorner p \urcorner^{[1]}}{p \Rightarrow p} \Rightarrow \text{-intro}^{[1]}$$

`[1] p [assumption]` declares the discharge binder. `p [from 1]` marks every leaf use. `[=> intro from 1]` discharges it.

$\vee$ -elim / case analysis (ternary — disjunction + two cases):

You write:

```
PROOF:
(p lor q) => r [=> intro from 1]
[1] p lor q [assumption]
:: r [lor elim from 1]
:: p lor q [from 1]
case p:
:: r [...derive r from p...]
case q:
:: r [...derive r from q...]
```

You get:

$$\frac{p \vee q \quad \frac{\ulcorner p \urcorner^{[1]}}{r} \quad \frac{\ulcorner q \urcorner^{[1]}}{r}}{r} \vee \text{-elim}$$

lor elim from N discharges the disjunction's assumption `[N]`; inside each **case X:** block, the case formula is referenced as **X [from N]**.

Relational Database Notation

Primary key annotation.

Mark a primary-key attribute with **pk** in a named schema body. The annotation sits outside the Z box so fuzz type-checks cleanly.

You write:

```
schema Book
  pk bookId : BookId
  isbn      : ISBN
  pages     : N
end
```

You get:

Book

bookId : *BookId*
isbn : *ISBN*
pages : $\mathbb{N}$

$$\text{PK}(\text{Book}) = \{bookId\}$$

Composite: two **pk** lines $\Rightarrow \text{PK}(S) = \{a, b\}$. Comma-separated names on one **pk** line also work. **pk** is rejected in **axdef/gendef**/inline schema text.

Relational algebra.

sigma [p](R)	Restrict _p (R) — restrict
pi [A,B](R)	Project _{A,B} (R) — project
R [B/A]	<i>R</i> [<i>B/A</i>] — rename (Z RM §3.11, NEW first)
R join S	Join(<i>R</i> , <i>S</i>) — natural join
R join [p] S	Join _p (<i>R</i> , <i>S</i>) — theta-join
R div S	<i>R</i> div <i>S</i> — division

sigma/pi bind at atom level. **R[B/A]** postfix rename requires relational context (inside **sigma/pi** arg or **join/div** right operand). **join** and **div** are infix at cross-product precedence.

Fuzz note: algebra expressions emit outside any Z environment — fuzz silently skips them. Do not wrap algebra inside **zed/axdef/schema** blocks.

FK predicate form.

Foreign-key constraints are expressed as axdef predicates:

You write:

```
axdef
where
  (R, S) |-> {a |-> b} elem FK
end
```

GROUP & UNGROUP.

Date’s nested-relation operators.

R group ({A,B} as x)	<i>R</i> GROUP({ <i>A</i> , <i>B</i> } AS <i>x</i>)
R ungroup x	<i>R</i> UNGROUP <i>x</i>

You write:

```
Members group ({name,email} as contact)
Members ungroup contact
```

You get:

Members GROUP ({*name*,*email*} AS *contact*)

Members UNGROUP *contact*

Fuzz note: GROUP/UNGROUP emit outside any Z environment. Do not place them inside **zed/axdef/schema** blocks.

GROUP aggregator form.

The aggregator keywords are **Count**, **Sum**, **Avg**, **Min**, **Max**, **Median**. Each takes one attribute name and an alias. The regroup form ({...} as alias) and the aggregator form are mutually exclusive in a single **group** expression.

R group (Count(x) as n)	<i>R</i> Group(Count(<i>x</i>) as <i>n</i>)
R group (Sum(x) as n)	<i>R</i> Group(Sum(<i>x</i>) as <i>n</i>)
R group (Avg(x) as n)	<i>R</i> Group(Avg(<i>x</i>) as <i>n</i>)
R group (Min(x) as n)	<i>R</i> Group(Min(<i>x</i>) as <i>n</i>)
R group (Max(x) as n)	<i>R</i> Group(Max(<i>x</i>) as <i>n</i>)
R group (Median(x) as n)	<i>R</i> Group(Median(<i>x</i>) as <i>n</i>)

You write:

```
Sales group (Count(orderId) as orderCount,
             Sum(amount) as totalRevenue)
```

You get:

Sales Group(Count(*orderId*) as *orderCount*,
Sum(*amount*) as *totalRevenue*)

Multiple aggregators are comma-separated. Each accepts exactly one attribute identifier — no nested aggregators, no compound expressions.

Quick reference.

pk a : T	primary key in schema
pk a, b : T	composite pk
$\sigma[p](R)$	restrict
$\pi[A](R)$	project
R[B/A]	rename attr (NEW/OLD)
R join S	natural join ($\text{Join}(R, S)$)
R div S	division
{ a == e }	binding bracket
{S; C P . E}	multi-typed set comp
R group ({A} as x)	regroup attrs
R group (Count(a) as n)	aggregate
R ungroup x	ungroup attr